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# Data Technician

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## Day 1: Task 1

Please complete the below boxes on commons laws and regulations that must be followed when working with customers data, use the below bulleted list to support your answers.

- What is it
- Why is it important
- Provide a real-world example of how you can follow it
- How does it impact working with data
- What could happen if you breached it

| | |
|---|---|
| **Data Protection Act** | **Data Protection (Data Protection Act / Data Protection Laws)** |

| | |
|---|---|
| | **What is it Data Protection?**<br><br>Is the collective name of laws and principles (like the UK Data Protection Act and GDPR) that control the way organisations collect, store, utilize and share personal data of individuals.<br><br>**Why is it important?**<br><br>It ensures the right of every individual to have their privacy protected and to know that their personal information is being processed in a fair, legal and secure manner. Besides, it also helps businesses to gain and maintain customers' reliance and prevent mishandling of delicate data.<br><br>**Provide a real-world example of how you can follow it**<br><br>An example of this would be the collection of only the necessary personal data for the service, its secure storage (e.g. encrypted databases), and the restriction of access to it to the staff with proper authorization.<br><br>**How does it impact working with data?**<br><br>You will only be allowed to use data that is necessary for your role.<br><br>Personal data must be stored in a safe and confidential manner.<br><br>Data has to be accurate and current.<br><br>Data can be kept for no longer than the period it is needed.<br><br>The transfer of data is only allowed if there is a valid legal reason.<br><br>**What could happen if you breached it?**<br><br>Financial penalties or fines.<br><br>Lawsuit from affected persons.<br><br>Disciplinary action or termination of employment.<br><br>Negative publicity and loss of customer loyalty. |
| **GDPR** | **GDPR (Regulation on General Data Protection)**<br><br>**What is it?**<br><br>GDPR is a data protection law that dictates how organisations should collect, store, process, or share personal information about individuals.<br><br>**Why is it important?** |

| | |
|---|---|
| | It safeguards individuals' right to privacy and ensures a company processes personal information lawfully, securely, and with transparency.<br><br>**Example of how to apply it in reality**<br><br>Encrypt customer databases.<br><br>Limit access to only authorized personnel.<br><br>Delete personal data once it is deemed unnecessary for business purposes.<br><br>**How it affects the process of working with data**<br><br>You should only store data which you require.<br><br>Data needs to be accurate, secure, and up to date.<br><br>Access to customer data must controlled and recorded.<br><br>Data cannot be shared without a valid legal basis.<br><br>**What might happen in case a breach occurred?**<br><br>Serious financial fines (up to 17.5 million or 4% of total company global turnover).<br><br>Litigation by Customers.<br><br>Loss of Trust and Reputational Harm. |
| **Freedom of Information Act** | **Freedom of Information Act (FOIA)**<br><br>**What is it?**<br><br>The Freedom of Information Act allows citizens to request information from public sector organisations.<br><br>**Why is it important?** |

| | |
|---|---|
| | It ensures that public bodies are transparent about their operations and decision-making processes.<br><br>**Real-world example of how to follow it**<br><br>In case a citizen seeks information on public expenditure, the agency should deliver the information within the time allowed by law unless it is covered by an exception.<br><br>**How does it impact working with data**<br><br>Data needs to be stored in a very organized manner so that it can be easily retrieved.<br><br>Before any personal or sensitive data is disclosed, it should be reviewed very carefully.<br><br>People working in data handling should be well trained on how to release information and how not to.<br><br>**What could happen if you breached it**<br><br>Legal actions by the Information Commissioner's Office (ICO).<br><br>Fines or official warnings.<br><br>Loss of public trust and damage to the organization's reputation. |
| **Computer Misuse Act** | **Computer Misuse Act**<br><br>**What is it The Computer Misuse Act?**<br><br>It prohibits unauthorized access to computer systems or data and destruction or alteration of systems or data.<br><br>**Why is it important?**<br><br>It has a paramount support in the fight against hacking, data theft, and cybercrime.<br><br>**Real-world example of how to follow it**<br><br>Only accessing the systems you have permission to access, utilizing strong passwords, and refrain from any action that might compromise the security measures.<br><br>**How does it impact working with data**<br><br>You should have access to the data only that is necessary for your job. |

No security measures should be overridden.

All suspicious access or activities must be reported.

**What could happen if you breached it**

Criminal prosecution.

Fines or prison sentences.

Termination of contract.

Severe loss of reputation.

## Day 2: Task 1

Please research and complete the following tasks within the retail-sales_dataset.xlsx document, paste a print screen into the provided boxes below:

1. In the sheet 'retail_sales_dataset' add all available data between columns A –J into a 'table'
2. Using the 'sort' function, sort 'Age' to 'largest to smallest'
3. Using the 'SUM' function, show me the commission total in cell 'L10'
4. Using the 'AVERAGE' function, show me the average commission in cell 'L11'

**Print screen 1**

| Transaction ID | Date | Customer ID | Gender | Age | Product Category | Quantity | Price per Unit | Total Sales | Commission |
|---|---|---|---|---|---|---|---|---|---|
| 74 | 11/22/2023 | CUST074 | Female | 18 | Beauty | 4 | $500.00 | $2,000.00 | $30.00 |
| 169 | 11/17/2023 | CUST169 | Male | 18 | Beauty | 3 | $500.00 | $1,500.00 | $22.50 |
| 283 | 5/8/2023 | CUST283 | Female | 18 | Electronics | 1 | $500.00 | $500.00 | $7.50 |
| 461 | 3/25/2023 | CUST461 | Female | 18 | Beauty | 2 | $500.00 | $1,000.00 | $15.00 |
| 595 | 11/9/2023 | CUST595 | Female | 18 | Clothing | 4 | $500.00 | $2,000.00 | $30.00 |
| 714 | 2/12/2023 | CUST714 | Female | 18 | Clothing | 1 | $500.00 | $500.00 | $7.50 |
| 16 | 2/17/2023 | CUST016 | Male | 19 | Clothing | 3 | $500.00 | $1,500.00 | $22.50 |
| 117 | 3/15/2023 | CUST117 | Male | 19 | Electronics | 2 | $500.00 | $1,000.00 | $15.00 |
| 257 | 2/19/2023 | CUST257 | Male | 19 | Beauty | 4 | $500.00 | $2,000.00 | $30.00 |
| 364 | 8/23/2023 | CUST364 | Female | 19 | Beauty | 1 | $500.00 | $500.00 | $7.50 |
| 412 | 9/16/2023 | CUST412 | Female | 19 | Electronics | 4 | $500.00 | $2,000.00 | $30.00 |
| 533 | 11/16/2023 | CUST533 | Male | 19 | Electronics | 3 | $500.00 | $1,500.00 | $22.50 |
| 677 | 10/27/2023 | CUST677 | Female | 19 | Beauty | 3 | $500.00 | $1,500.00 | $22.50 |

| | |
|---|---|
| **Print screen 2** | |

| E |
|---|
| Age |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |
| 64 |

| | |
|---|---|
| **Print screen 3** | |

| | |
|---|---|
| Sum_Commission | =SUM(J2:J1001) |
| Sum_Commission | $6,840.00 |

| | |
|---|---|
| **Print screen 4** | |

| | |
|---|---|
| Average_Commission | =AVERAGE(J2:J1001) |
| Average_Commission | $6.84 |

## Day 2: Task 2

Please research and complete the following tasks within the retail-sales_dataset.xlsx document, paste print screens into the provided box below:

| Student name | English | Mathematic: | Science | Average | Highest score |
|---|---|---|---|---|---|
| Carol | 75 | 85 | 85 | | |
| Ted | 80 | 75 | 90 | | |
| Khan | 85 | 75 | 80 | | |
| Harry | 80 | 70 | 80 | | |
| Sarah | 80 | 70 | 80 | | |
| John | 65 | 80 | 70 | | |
| Linda | 90 | 50 | 70 | | |
| Edward | 55 | 80 | 60 | | |
| Mary | 55 | 70 | 65 | | |
| Thomas | 55 | 30 | 65 | | |

Task

1) Apply filter and sorting to show the best students in each subject.
2) Calculate the average for all students and fill into Column E. (Use formula)
3) Using the =MAX fucntion, tell me what the students highest score was in column F.
4) Apply filter and sorting to show the best student in this classroom by average.
5) Apply filter and sorting to show the best student in this classroom by highest score.
6) Use conditional formatting to clearly identify the highest and lowest average scores

**Print screen 1**

| B | C | D | E | F | E |
|---|---|---|---|---|---|
| English | Mathema | Science | Averag | Highest sco | Averag |
| 90 | 85 | 90 | 50 | 65 | 82 |
| 85 | 80 | 85 | 63 | 70 | 82 |
| 80 | 80 | 80 | 65 | 80 | 80 |
| 80 | 75 | 80 | 70 | 80 | 77 |
| 80 | 75 | 80 | 72 | 80 | 77 |
| 75 | 70 | 70 | 77 | 80 | 72 |
| 65 | 70 | 70 | 77 | 85 | 70 |
| 55 | 70 | 65 | 80 | 85 | 65 |
| 55 | 50 | 65 | 82 | 90 | 63 |
| 55 | 30 | 60 | 82 | 90 | 50 |

| F | E |
|---|---|
| Highest sco | Averag |
| 90 | 82 |
| 90 | 70 |
| 85 | 82 |
| 85 | 80 |
| 80 | 77 |
| 80 | 77 |
| 80 | 72 |
| 80 | 65 |
| 70 | 63 |
| 65 | 50 |

## Day 2: Task 3

Using the skills developed today, have some fun with the data set you have imported. Paste your work below and enjoy!

**Print screen 1**

## Day 3: Task 1

Please download the dataset 'Day_3_Task_1_Bike_Sales_Pivot_Lab.xlsx' and the lab instructions.

Do not worry if you do not complete the lab, just working with data and playing with the pivot table will be good experience.

Please paste your final pivot table below and complete the reflection questions:

| | |
|---|---|
| **Print screen 1** | |
| **In which markets does Germany have customers?** | |
| **What country has sales in all markets?** | |
| **What are the most profitable markets by country, age group, and gender?** | |
| **Any other findings?** | |

## Day 3: Task 2

The dataset below tracks the sales performance of different products in various counties in England. Please paste the dataset into a blank Excel workbook. Your task is to:

- **Create a Pivot Table** to summarise the data by county and product.
- **Use the SWITCH function** to categorise products based on their sales volume.

### *Dataset:*

| County | Product | Sales Volume |
|---|---|---|
| Yorkshire | Laptops | 500 |
| Yorkshire | Smartphones | 200 |
| Cornwall | Laptops | 700 |
| Cornwall | Printers | 400 |
| Lancashire | Smartphones | 150 |
| Lancashire | Laptops | 600 |
| Essex | Printers | 800 |
| Essex | Smartphones | 300 |
| Durham | Laptops | 250 |
| Durham | Printers | 300 |
| Greater Manchester | Smartphones | 600 |
| Greater Manchester | Laptops | 400 |

### *Step 1: Create a Pivot Table*

- Select the dataset (columns A to C).
- Insert a Pivot Table to summarise the data by **County** in the rows and **Products** in the columns. Use **Sales Volume** as the value to be summarised.

### *Step 2: Use the SWITCH Function*

In a new column next to your data, use the SWITCH function to categorise products based on **Sales Volume** as follows:

- For sales greater than 600: **"High"**
- For sales between 300 and 600: **"Medium"**
- For sales less than 300: **"Low"**

**SWITCH Function Example**:

```
=SWITCH(TRUE, C2 > 600, "High", C2 >= 300, "Medium", "Low")
```

- Apply this formula to each row, and check if the products are categorised correctly.

***Submission:***

- A completed Pivot Table summarising sales by county and product.
- A new column in the dataset categorising products by sales volume using the SWITCH function.
    - Please paste your completed work below

| | |
|---|---|
| **Print screen 1** | |

## Day 3: Task 3

Please download the dataset 'Day_3_Task_3_Bike_Sales_Visualisations_Lab.xlsx' and the the lab instructions. Do not worry if you do not complete the lab, just working with data and playing with the charts will be good experience.

Please paste your results below:

| | |
|---|---|
| **Print screen 1** | |

## Day 4: Task 1

You have been asked to deliver your analysis findings to the board of directors, with your analysis you have identified that customers are leaving your company at the 12-month point, this is typically when they receive their renewal price.

Conduct research and complete the below questions:

| | |
|---|---|
| **How would you prepare for the delivery?** | |
| **What tools would you use for the delivery?** | |
| **What is prospecting and why would you complete this before your delivery?** | |
| **Tell me best practices for public speaking and providing updates to senior leaders** | |

| | |
|---|---|
| What will you show the board in your delivery? | |
| How will you articulate the changes that are needed? | |
| Provide a list of online resources and videos that will support your preparation for public speaking | |
| Evaluate tools that provide visualisation.<br><br>Tell me what they are.<br><br>Tell me what you would choose when delivering your presentation and why | |

# Course Notes

It is recommended to take notes from the course, use the space below to do so, or use the revision guide shared with the class:

We have included a range of additional links to further resources and information that you may find useful, these can be found within your revision guide.

**END OF WORKBOOK**

**Please check through your work thoroughly before submitting and update the table of contents if required.**

**Please send your completed work booklet to your trainer.**